



## INTELLECTUAL PROPERTY

# Patents and Trademarks

Twenty-nine issued patents, with pending applications and trademarks, spanning operative safety and efficiency, mixed-reality implant placement, revision hip and knee fixation, soft-tissue tensioning, bacteriophage delivery, and dormant-infection detection. The portfolio is held by Stanford University and the five companies founded by Derek F. Amanatullah, M.D., Ph.D.

## Assigned to Stanford University 6

### **System and Method for Detecting, Characterizing, and Preventing Unintentional Tissue Damage During a Surgical Procedure**

#62/991,546, #17/206,012 (allowed), PCT-US21/23028

### **Method for Monitoring Object Flow within a Surgical Space During a Surgery**

#62/875,283, #16/533,632, #16/932,607, #17/962,316, #18/085,509, #11,583,360, #11,666,409, PCT-US20/42661

### **Controlled Release of Bacteriophage to Treat Implant Infections**

#63/024,963, #18/552,406, #EU4314120, #PCT-US2022/071351

### **Methods for Detecting a Dormant Infection**

#63/677,070, PCT-US25/039107

### **Method for Detecting Subclinical Bacterial Infections**

#63/848,642

### **Method and Device for Detecting Dormant and Active Infection**

#63/6965,953

## Assigned to Arthrology Consulting, LLC 7

### **Design of Repetitive-A Jewelry Piece**

#29/594,948, #D873701

### **Cool-cut Cast Saw Blade**

#14/102,403, #9,468,570

### **Modular Total Knee Arthroplasty System and Method**

#62/318,684, #15/477,578, #16/209,888, #16/833,194, #17/661,629, #18/908,005, #10,182,830, #10,639,047, #11,344,320, #12,114,871

**Expandable Augment System for Acetabular Cup**

#17/455,893, #18/590,344, #19/229,431, #11,944,544, #12,343,259

**Off-center Liner for Acetabular Cup**

#17/455,130, #18/511,283, #11,857,423

**Distractor Tool and System for Balancing a Hip Joint in Hip Joint Replacement Surgery**

#63/896,174

**Phase-shifted Femoral Heads**

#63/896,154

## Under Exclusive License to Medacta 4

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**Method for Augmenting a Surgical Field with Virtual Guidance Content**

#62/328,330, #62/612,895, #15/499,046, #15/594,623, #16/588,926, #10,194,990, PCT-US18/29506

**Method for Augmenting a Surgical Field with Virtual Guidance and Tracking and Adapting to Deviations from a Surgical Plan**

#16/238,500, #16/238,504, #17/575,545, #10,806,518, #10,813,700, #11,253,321, PCT-US19/12111

**Method for Detecting and Tracking Deviations from a Surgical Plan**

#62/612,901

**Method for Patient Registration and Planning for Surgical Customization, Visualization, and Evaluation, of Intra Operative Deviation and Post Execution Accuracy**

#62/363,022

## Assigned to Arthrology Designs, LLC (DBA: PlantarTech) 4

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**Trademark PlantarTech, Class 10: Splints for Feet**

#7,249,638

**Trademark , Class 10: Splints for Feet**

#6,687,181

**Dynamic Hallux Tension Device for Plantar Fasciitis**

#13/745768, #15/686,515, #9,320,637, #9,750,631, #10,758,395

**Foot Splint**

#29/788,488, #D1082015

## Assigned to Knimble Designs, Inc. 5

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**Methods for Generating Intraoperative Surgical Guidance during Joint Arthroplasty Based on Dynamic Ligament Tension**

#62/853,600, #16/886,413, #17/947,670, #18/677,753, #11,478,302, #12,023,104

**System for Total Knee Replacement**

#63/108,243, #17/515,205, #18/419,350, #11,911,050

**Method and Apparatus for Mid-flexion Balancing during Knee Arthroplasty**

#18/127,250, #18/242,612, #18/503,600, #11,786,262, #11,849,960, PCT-US24/20892, WO-2024/206057, AUS-2024/254804, CAN-3,286,912, JAP-2025/556580, EPO-24781587.1

**System for Identifying Positions of an Artificial Tibial Component during Total Knee Arthroplasty**

#63/523,303, #18/755,524, #12,648,861

**Method and Apparatuses for Performing a Knee Arthroplasty**

#63/575,092, #63/672,332, #19/170,618, #19/263,224 (allowed), #19/263,650 (allowed), #19/264,495, PCT-US25/23209

## Assigned to Zooly Labs, Inc. 2

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**System for Balancing a Hip During a Hip Arthroplasty**

#63/321,013, #18/123,177, PCT-US23/15552

**Combined Sensing and Adjustment Tool for Hip Arthroplasty Balancing**

#63/693,135, #19/325,361

## Assigned to nSight Surgical, Inc. 4

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**Surgical Imaging System and Method**

#63/285,466, #63/285,467, #18/074,408

**Multicamera Tracking Set-up: Visual Tracking of Lap Sponges, Needles Across OR**

#63/347,144

**Method for Tracking Objects within a Surgical Space**

#63/429,509

**Method for Predicting Consumption of Surgical Consumables During Surgical Operation**

#63/529,117, #18/786,053